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A Multi-Device Multi-Position (MDMP) Smartphone Sensing Dataset for Human Activity Recognition

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Abstract. Human Activity Recognition (HAR) using smartphone sensors has gained significant attention due to its potential applications in healthcare, fitness tracking, and context-aware computing. While deep learning models, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, have demonstrated superior performance in handling sequential sensor data, the impact of smartphone placement position on recognition accuracy remains under-explored. This study systematically investigates the influence of three common smartphone positions—Handheld (HH), Shirt Pocket (SPT), and Right Pant Pocket (RPT)—on HAR performance. A comprehensive, position independent dataset was collected from multiple users using different devices, capturing static (sitting, standing, lying) and dynamic (walking, jogging) activities via smartphone accelerometer across these positions¹. We evaluated state of the art deep learning models such as RNN and LSTM, to determine their robustness for positional variations. The results indicate that LSTM outperforms RNN in most of the activity recognition, with the highest accuracy (97%) observed when the smartphone is placed in the RPT position. These findings highlight the significance of smartphone placement in HAR and underscore the need for position-aware modeling to enhance system reliability.

Keywords: Deep Learning · Activity Recognition · Smartphone sensor

1 Introduction

In recent years, human activity recognition using smartphone sensors has become increasingly prevalent, driven by the proliferation of mobile devices and their ubiquitous integration into everyday life. This technology holds immense promise for a wide range of applications, including healthcare monitoring, fitness tracking, context-aware computing, and human-computer interaction [1],[2]. Smartphones, equipped with various embedded sensors, offer a unique blend of

¹ https://drive.google.com/drive/folders/18bkBK7PxBggN_VyK6z8En9pT2coJcWTX

2 Indradeep Pal , Anindita Saha, and Chandreyee Chowdhury

sensing, communication, and computation capabilities, enabling the development of pervasive behavior monitoring applications for everyday environments. Over the past decade, machine learning-based solutions have successfully recognized human activities [3][4][5], but there have been challenges in implementing them such as limited training data for multiclass activity recognition, difficulty in recognizing complex features using manual feature extraction. Additionally, the dynamic nature of activities and motion variations in different scenarios complicates accurate recognition [6]. Fig 1 exhibits the basic working principle of HAR with Machine Learning (ML) model.

Recent advancements in deep learning models have demonstrated significant and promising performance improvements across various benchmark datasets used in machine learning solutions [7][8][9]. These models can streamline the data preprocessing and feature extraction phases. In smartphone-based activity recognition, deep learning models like RNNs and LSTMs are crucial for capturing temporal dependencies in sequential sensor data. Traditional methods struggle with these complex patterns, leading to lower accuracy. RNNs and LSTMs excel in handling long-range dependencies, with LSTMs addressing the vanishing gradient problem, making them ideal for activities that unfold over extended periods [10][11][12][13]. These models significantly enhance the accuracy and robustness of activity recognition systems, enabling more advanced, context-aware applications. Further, they enhance the generalization performance and robustness of the models by reducing the dependency on expert knowledge for feature engineering, accurately recognize the temporal dynamics of features, shorten testing times for image-based recognition, and maintain high performance even with weakly labeled data [14].

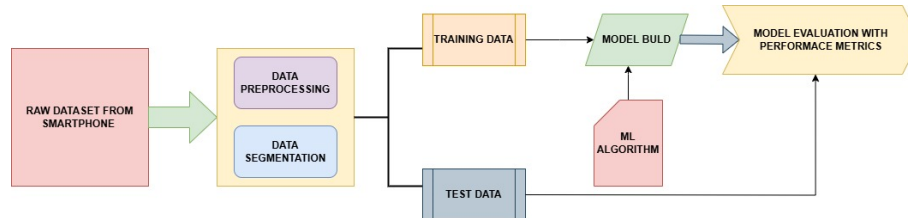


Fig. 1: Basic Framework of HAR

However, the performance of these models is influenced by smartphone placement during data collection, as suggested in [15][16][17]. Optimal sensor placement depends on the specific activity, highlighting the importance of position awareness for accelerometer-based detection. Despite significant research in Human Activity Recognition (HAR) using smartphones, a notable gap persists in the evaluation of recognition performance concerning the position independence of the smartphone. Existing publicly available datasets (such as, UCI-HAR [18], WISDM [19], MHEALTH [20]) incorporate variations across users and devices;

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however, they lack a focused consideration of position-specific data collection, particularly in natural, real-world smartphone placements such as in hand, shirt pocket, and pant pocket. These placements are crucial, as they mirror common user behavior during daily activities. To bridge this gap, we propose a position-wise separated dataset, specifically curated to evaluate the impact of smartphone placement on activity recognition. While classical machine learning techniques have been employed in similar domains [21][22], there remains a lack of systematic exploration using deep learning approaches, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which are well-suited for temporal sequence modeling in sensor-based HAR. Addressing this issue is essential to developing position-invariant and reliable HAR models with higher real-world applicability. This work is therefore motivated by the need to analyze the influence of the position independence of a smartphone orientation on recognition accuracy, leveraging deep learning models on a real-time dataset collected from multiple users.

The main contribution in this paper are as follows:

- A comprehensive, multi-device multi-position (MDMP) dataset has been proposed, which is collected from different users through different devices, performing both static activities and dynamic activities, using smartphone accelerometer sensor.
- The impact of the position of the Smartphone on the recognition performance has been thoroughly investigated through deep learning models RNN and LSTM.

The work is organized as follows. Section 2 presents the proposed dataset design and methodology for analysis. Experimental setup and results are discussed in Section 3 while Section 4 concludes the work.

2 Methodological Framework

In this section, we delineate the proposed methodology designed to address the research objectives outlined earlier. This section provides a comprehensive overview of the methods and procedures employed aiming to achieve the desired outcomes efficiently.

2.1 Dataset Description

To facilitate a comprehensive understanding of our research, we present a detailed description of the dataset used in this work. Effective activity recognition depends heavily on the quality of data collection. For this work, data was collected using accelerometers embedded in smartphones, carried by five users (2 Females and 3 Males) between the age of 18-25 years, with five different device configurations. A standard Android app, G-Sensor Logger ² developed by

² https://play.google.com/store/apps/details?id=com.peterhohsy.gsensor_debug&hl=en_IN.

4 Indradeep Pal , Anindita Saha, and Chandreyee Chowdhury

Kewlsoft, was used to capture the three-dimensional acceleration data, which includes measurements of the device's movement along the x, y, and z axes. The users performed two dynamic activities like "Walking", "Jogging" and three static activities such as "Standing", "Sitting" and "Laying". The Smartphones were kept at three different positions, namely "Handheld" (HH), "Shirt Pocket" (SPT) and "Right Pant Pocket" (RPT) during each of these activities for each user. The collected accelerometer data, was consolidated into three CSV files as per their respective positions and along with the x, y, and z-axes, an activity column is also augmented with the csv for further processing. The data set therefore comprised of 15 lakh instances along with 4 columns, namely, 'X axis', 'Y axis', 'Z axis', and 'Activity', respectively. A comprehensive description of the dataset is shown in Table 1.

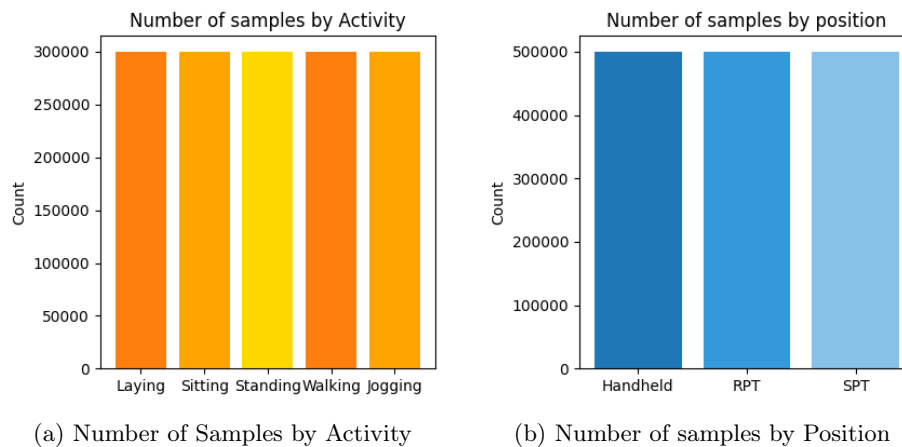
Table 1: Dataset Description

Device Type	Smartphone
Type of dataset	Collected by users in real time
Number of Users	5
Device used	Samsung Galaxy M11, Poco C3, Redmi 12C, Samsung A14, Vivo V27
Number of Features	3
Sensor	Accelerometer
Activities Performed	Laying, Sitting, Standing, Walking and Jogging
Position	Handheld (HH), Shirt Pant Pocket (SPT), Right Pant Pocket (RPT)
Sampling Rate	50 Hz

2.2 Proposed Methodology

The primary objective of this study is to rigorously evaluate the performance of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks in analyzing sensor data from smartphones placed in three distinct positions. These architectures are leveraged for their ability to capture temporal dependencies and complex activity patterns. For experimentation, the dataset is categorized based on smartphone placement in three distinct orientations—HH, SPT, and RPT—resulting in three separate CSV files. Each CSV file contains sensor data from all five users across all activities. Specifically, CSV1 includes HH data from all users for all activities, CSV2 contains SPT data from all users for all activities, and CSV3 comprises RPT data from all users for all activities. This structured organization ensures a comprehensive evaluation of activity recognition across different smartphone positions. Figure 2a presents a graphical representation of the sample count for each activity, while Figure 2b illustrates the distribution of samples across the three positions (HH, SPT, and RPT). As evident from the figures, we present a well balanced dataset with

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uniform sample count across the classes. Class imbalance problem commonly present in many state-of-the art HAR datasets could not be observed here.

As part of the preprocessing pipeline, the dataset is transformed from a 2D format (rows and columns) into a 3D tensor (samples, timesteps, features), facilitating the sequential data processing required for RNNs and LSTMs. A sliding window approach is employed, with a window size of 50 timesteps and a step size of 10, ensuring an effective representation of temporal patterns. Initially, RNN is applied separately to each position based dataset, followed by LSTM, with performance evaluations and comparative results discussed in the subsequent section.

3 Experiments on RNN and LSTM

As discussed, two state-of-the-art deep learning (DL) models were applied independently to each of the three position-specific datasets, corresponding to the different smartphone placements. RNN is applied separately to each position-based dataset (HH, SPT, and RPT), and accuracy is evaluated for each activity. Figure 3a presents the results for the HH dataset, where dynamic activities such as, Walking and Jogging are well recognized. However, among static activities, Standing is less accurately classified compared to others, indicating potential challenges in distinguishing subtle variations in posture when the smartphone is handheld. This could be attributed to minor movements or instability in the device's placement during data collection.

Upon repositioning the smartphone to SPT, as shown in the Fig 3b, the model demonstrates a notable improvement in recognizing both static and dynamic activities, along with the accuracy for Standing increasing significantly from 72.18% to 92.83%. This suggests that shifting from HH to SPT leads to better sensor stability, thereby enhancing model performance. Accelerometer be-

6 Indradeep Pal , Anindita Saha, and Chandreyee Chowdhury

ing a contact sensor, could sense the body acceleration better when kept in shirt pocket.

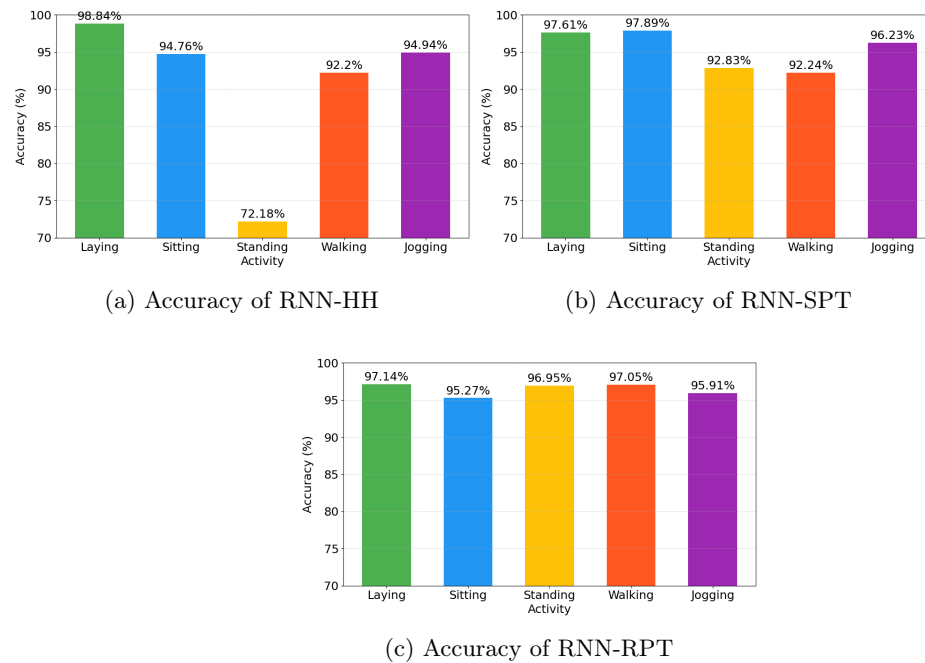


Fig. 3: Accuracy of RNN model on all three positions of the Smartphone for all activities

Further repositioning to RPT results in even higher accuracy, with Standing improving from 92.83% to 96.95%. This happens because our center of gravity lies near the pelvic region. This substantial increase highlights that RPT placement provides the most stable and consistent sensor readings, leading to more reliable activity recognition. Additionally, the classification accuracy for other activities remains largely consistent across different positions, with no significant fluctuations. This suggests that while static activities like standing are more position-dependent with RNN, **dynamic activities such as Walking and Jogging are consistently well recognized**, regardless of smartphone placement.

Similar experiments were conducted using LSTM as the deep learning model to evaluate its effectiveness in HAR across different smartphone positions as shown in the Fig 4a, 4b and 4c . The results indicate that when the smartphone is positioned in the HH orientation, the model exhibits challenges in accurately identifying static activities, particularly Standing, just like RNN, achieving an accuracy of 75.48%. With the smartphone repositioned to the SPT orientation, the accuracy of the Standing activity significantly improved to 94.61%, marking

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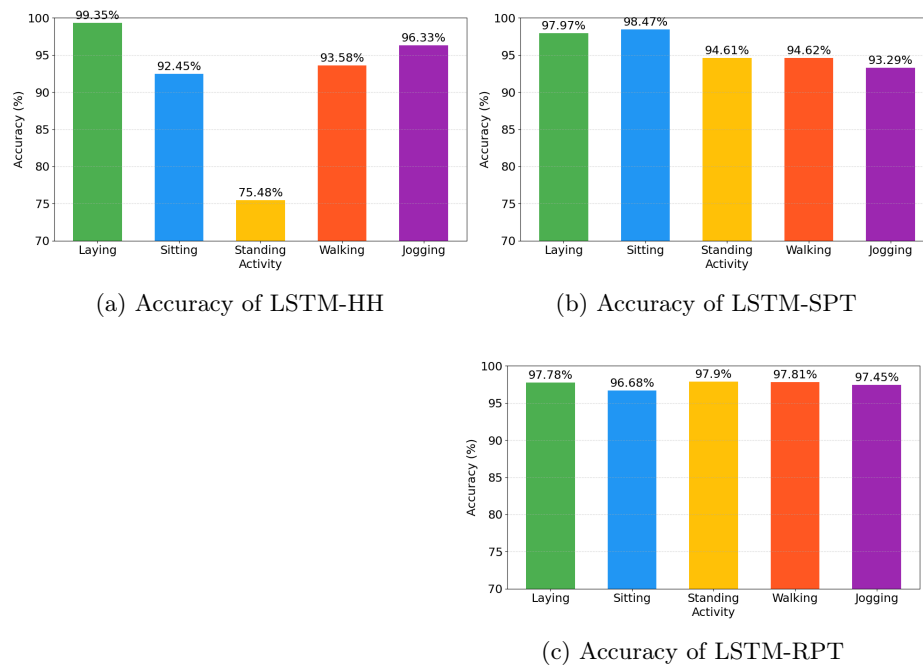


Fig. 4: Accuracy of LSTM model on all three positions of the Smartphone for all activities

a substantial enhancement compared to the HH position. This notable increase suggests that the placement of the device in SPT provides more stable and reliable sensor data, allowing the model to better capture and distinguish the characteristics of static activities.

For the RPT position, the accuracy of the same Standing activity improved to 97.9%, making it the most optimal placement for activity recognition. This improvement highlights the critical role of smartphone positioning in enhancing model performance, likely due to improved sensor stability and reduced motion artifacts. However, for other activities, the accuracy remained relatively consistent across different smartphone placements, indicating that position primarily influences specific activities rather than overall classification performance. Table 2 summarizes the results in a nutshell.

3.1 Comparison of Performance of RNN and LSTM based on Smartphone Position

The comparative analysis of RNN and LSTM across different smartphone positions (HH, SPT, and RPT) reveals that LSTM consistently outperforms RNN in most cases, demonstrating its superior ability to capture complex temporal dependencies in sequential data, as shown in Table 2. However, the effectiveness

8 Indradeep Pal , Anindita Saha, and Chandreyee Chowdhury

of both models is significantly influenced by the placement of the smartphone, with the RPT position emerging as the most optimal for activity recognition. For static activities, particularly Laying and Sitting, LSTM achieves slightly higher accuracy across all smartphone placements. In Laying (HH), LSTM records 99.35% compared to 98.84% for RNN, while in Sitting (SPT and RPT), LSTM outperforms RNN with 98.47% vs. 97.89% and 96.68% vs. 95.27%, respectively. However, in Sitting (HH), RNN slightly surpasses LSTM (94.76% vs. 92.45%), suggesting that for certain static activities, RNN can still remain competitive. The substantial increase in accuracy for Standing when transitioning from HH to

Table 2: Comparison of Accuracies for all positions and for all activities

Activity	Position	Accuracy of RNN	Accuracy of LSTM
Laying	HH	98.84	99.35
	SPT	97.61	97.97
	RPT	97.14	97.78
Sitting	HH	94.76	92.45
	SPT	97.89	98.47
	RPT	95.27	96.68
Standing	HH	72.18	75.48
	SPT	92.83	94.61
	RPT	96.95	97.90
Walking	HH	92.90	93.58
	SPT	92.24	94.62
	RPT	97.05	97.81
Jogging	HH	94.94	96.33
	SPT	96.23	93.29
	RPT	95.91	97.45

RPT indicates that RPT is the most effective placement for recognizing postural activities, likely due to enhanced sensor stability and reduced noise in the collected data. For dynamic activities like Walking and Jogging, both models exhibit high accuracy, though LSTM generally maintains an edge. For Jogging (HH), LSTM achieves 96.33%, surpassing RNN's 94.94%. However, Jogging (SPT) is an exception, where RNN records a higher accuracy of 96.23% compared to 93.29% for LSTM, suggesting that for certain dynamic movements, RNN may provide slightly better generalization. Despite this, LSTM remains superior for most dynamic activities across all smartphone positions.

A key observation is that the placement of the smartphone has a significant impact on classification performance, particularly for static activities. When the smartphone is in the HH position, both models struggle to recognize Standing accurately, with RNN achieving only 72.18% accuracy and LSTM performing slightly better at 75.48%. However, repositioning the smartphone to SPT leads to a major improvement in accuracy, with Standing increasing to 92.83% for RNN and 94.61% for LSTM. This suggests that the SPT position enhances

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model performance by providing more stable sensor readings. The best results are obtained when the smartphone is placed in the RPT position, where Standing reaches 96.95% (RNN) and 97.90% (LSTM), confirming that RPT is the optimal smartphone position for activity recognition. Additionally, the accuracy of most activities remains stable when transitioning from SPT to RPT, further reinforcing the reliability of RPT as the best placement. Overall, LSTM consistently outperforms RNN in both static and dynamic activity recognition, demonstrating its superior ability to model sequential dependencies. Table 3 summarizes the optimal position of the Smartphone with LSTM model for each activity.

Table 3: Ideal positions of smartphone with LSTM model

Activity	Accuracy (HH)	Accuracy (RPT)	Accuracy (SPT)	Recommended Position
Laying	99.35	97.78	97.97	Hold it in Hand
Sitting	92.45	96.68	98.47	Keep it in the Shirt Pocket
Standing	75.48	97.90	94.61	Keep in the Right Pocket
Walking	93.58	97.81	94.62	Keep in the Right Pocket
Jogging	96.33	97.45	93.29	Keep in the Right Pocket

4 Conclusion

This work presents a multi-device multi-position accelerometer dataset and explored the impact of smartphone placement on human activity recognition using deep learning models, specifically RNNs and LSTMs. Our findings demonstrate that smartphone position significantly influences recognition accuracy, with data collected from different placements yielding varying results. By analyzing a dataset collected from five users performing both static (sitting, standing, laying) and dynamic (walking, jogging) activities in three different smartphone positions (Handheld, Shirt Pocket, and Right Pant Pocket), we established that LSTM consistently outperforms RNN in classification accuracy. Moreover, the Right Pant Pocket (RPT) position yielded the highest accuracy across activities, highlighting its suitability for reliable sensor-based activity recognition. These insights emphasize the need to consider smartphone placement when designing robust HAR systems and pave the way for further research into optimizing deep learning models for real-world, context-aware applications.

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